

MLR Part1

Now let's look at an example concerning the number of species found on the various Galápagos Islands. There are 30 cases (Islands) and seven variables in the dataset. We start by reading the data into R and examining it:

```
load(url("https://raw.githubusercontent.com/cran/faraway/master/data/gala.rda"))
head(gala[, -2])
```

```
##           Species Area Elevation Nearest Scrutz Adjacent
## Baltra          58 25.09         346    0.6   0.6    1.84
## Bartolome        31  1.24          109    0.6  26.3   572.33
## Caldwell         3  0.21           114    2.8  58.7    0.78
## Champion        25  0.10           46    1.9  47.4    0.18
## Coamano          2  0.05           77    1.9   1.9   903.82
## Daphne.Major    18  0.34          119    8.0   8.0    1.84
```

The variables are Species — the number of species found on the island, Area — the area of the island (km²), Elevation — the highest elevation of the island (m), Nearest — the distance from the nearest island (km), Scrutz — the distance from Santa Cruz Island (km), Adjacent — the area of the adjacent island (km²). We have omitted the second column (which has the number of endemic species) because we shall not use this alternative response variable in this analysis.

```
lmod<-lm(Species~Area+Elevation+Nearest+Scrutz+Adjacent,data=gala)
summary(lmod)
```

```
##
## Call:
## lm(formula = Species ~ Area + Elevation + Nearest + Scrutz + Adjacent,
##     data = gala)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -111.679  -34.898   -7.862   33.460  182.584
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.068221   19.154198   0.369  0.715351
## Area        -0.023938    0.022422  -1.068  0.296318
## Elevation    0.319465    0.053663   5.953 3.82e-06 ***
## Nearest      0.009144    1.054136   0.009  0.993151
## Scrutz       -0.240524    0.215402  -1.117  0.275208
## Adjacent     -0.074805    0.017700  -4.226  0.000297 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 60.98 on 24 degrees of freedom
## Multiple R-squared:  0.7658, Adjusted R-squared:  0.7171
## F-statistic: 15.7 on 5 and 24 DF,  p-value: 6.838e-07
```

Lets extract X-Matrix

```
X<-model.matrix(lmod)
```

Lets extract Y-Matrix

```
Y<-gala$Species
```

Lets construct Hat matrix $(X^T X)^{-1} X^T$

```
H<-solve(t(X)%*(X))%*t(X)
```

Lets calculate β

```
Beta<-H%*Y
```

```
Beta
```

```
##           [,1]
## (Intercept)  7.068220709
## Area        -0.023938338
## Elevation    0.319464761
## Nearest      0.009143961
## Scrutz       -0.240524230
## Adjacent     -0.074804832
```

We can extract the regression quantities we need from the model object. Commonly used are residuals (), fitted (), df.residual () which gives the degrees of freedom, deviance () which gives the RSS and coef () which gives the $\hat{\beta}$. You can also extract other needed quantities by examining the model object and its summary:

We can estimate σ using the formula in the text above or extract it from the summary object:

```
names(lmod)
```

```
## [1] "coefficients" "residuals"      "effects"      "rank"
## [5] "fitted.values" "assign"         "qr"           "df.residual"
## [9] "xlevels"       "call"          "terms"        "model"
```

```
lmodsum<-summary(lmod)
```

```
names(lmodsum)
```

```
## [1] "call"          "terms"          "residuals"      "coefficients"
## [5] "aliases"       "sigma"          "df"             "r.squared"
## [9] "adj.r.squared" "fstatistic"     "cov.unscaled"
```

```
sqrt(deviance(lmod)/df.residual(lmod))
```

```
## [1] 60.97519
```

```
lmodsum$sigma
```

```
## [1] 60.97519
```

We can also extract $(X^T X)^{-1}$ and use it to compute the standard errors for the coefficients. (diag () returns the diagonal of a matrix):

```
xtxi<-lmodsum$cov.unscaled
```

```
sqrt(diag(xtxi))*lmodsum$sigma
```

```
## (Intercept)      Area      Elevation      Nearest      Scrutz      Adjacent
## 19.15419782  0.02242235  0.05366280  1.05413595  0.21540225  0.01770019
```

```
summary(lmod)
```

```
##
```

```
## Call:
## lm(formula = Species ~ Area + Elevation + Nearest + Scrutz + Adjacent,
##     data = gala)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -111.679  -34.898   -7.862   33.460  182.584
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.068221  19.154198   0.369 0.715351
## Area        -0.023938   0.022422  -1.068 0.296318
## Elevation    0.319465   0.053663   5.953 3.82e-06 ***
## Nearest      0.009144   1.054136   0.009 0.993151
## Scrutz       -0.240524   0.215402  -1.117 0.275208
## Adjacent     -0.074805   0.017700  -4.226 0.000297 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 60.98 on 24 degrees of freedom
## Multiple R-squared:  0.7658, Adjusted R-squared:  0.7171
## F-statistic: 15.7 on 5 and 24 DF,  p-value: 6.838e-07
```

```
lmodsum$coef[,2]
```

```
## (Intercept)      Area  Elevation  Nearest  Scrutz  Adjacent
## 19.15419782  0.02242235  0.05366280  1.05413595  0.21540225  0.01770019
```

Creating Dummy Variables

Bringing data into R

```
IID <- read.csv("/cloud/project/Data/Insurance Inovation Data.csv")
```

Using Libraries

it will create dummy variables for categorical variables. In this case, it will create two dummy variables but we need only one, you need to manually drop it or don't use it

```
library('fastDummies')
IID.Dat<-dummy_cols(IID, select_columns = 'Firm.Type')
IID.Dat[1:3,]
```

```
##      Y  X1 X2 Firm.Type Firm.Type_Mutual Firm.Type_Stock
## 1 17 151  0   Mutual              1              0
## 2 26  92  0   Mutual              1              0
## 3 21 175  0   Mutual              1              0
```

```
f<-lm(Y~X1+Firm.Type_Stock,data=IID.Dat)
summary(f)
```

```
##
## Call:
## lm(formula = Y ~ X1 + Firm.Type_Stock, data = IID.Dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -5.6915 -1.7036 -0.4385  1.9210  6.3406
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   33.874069   1.813858  18.675 9.15e-13 ***
## X1            -0.101742   0.008891 -11.443 2.07e-09 ***
## Firm.Type_Stock 8.055469   1.459106   5.521 3.74e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.221 on 17 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8827
## F-statistic: 72.5 on 2 and 17 DF,  p-value: 4.765e-09
```

using factor command

Factor command will automatically create a dummy variable. For variable selection, it is not recommended.

```
f<-lm(Y~X1+factor(Firm.Type),data=IID)
summary(f)
```

```
##
## Call:
## lm(formula = Y ~ X1 + factor(Firm.Type), data = IID)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.6915 -1.7036 -0.4385  1.9210  6.3406
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   33.874069   1.813858  18.675 9.15e-13 ***
## X1            -0.101742   0.008891 -11.443 2.07e-09 ***
## factor(Firm.Type)Stock 8.055469   1.459106   5.521 3.74e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.221 on 17 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8827
## F-statistic: 72.5 on 2 and 17 DF,  p-value: 4.765e-09
```

manually coding

This is the recommended approach

```
IID$X.Stock=I(IID$Firm.Type=="Stock")*1
IID$X.Stock
```

```
## [1] 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1
```

#alternatively

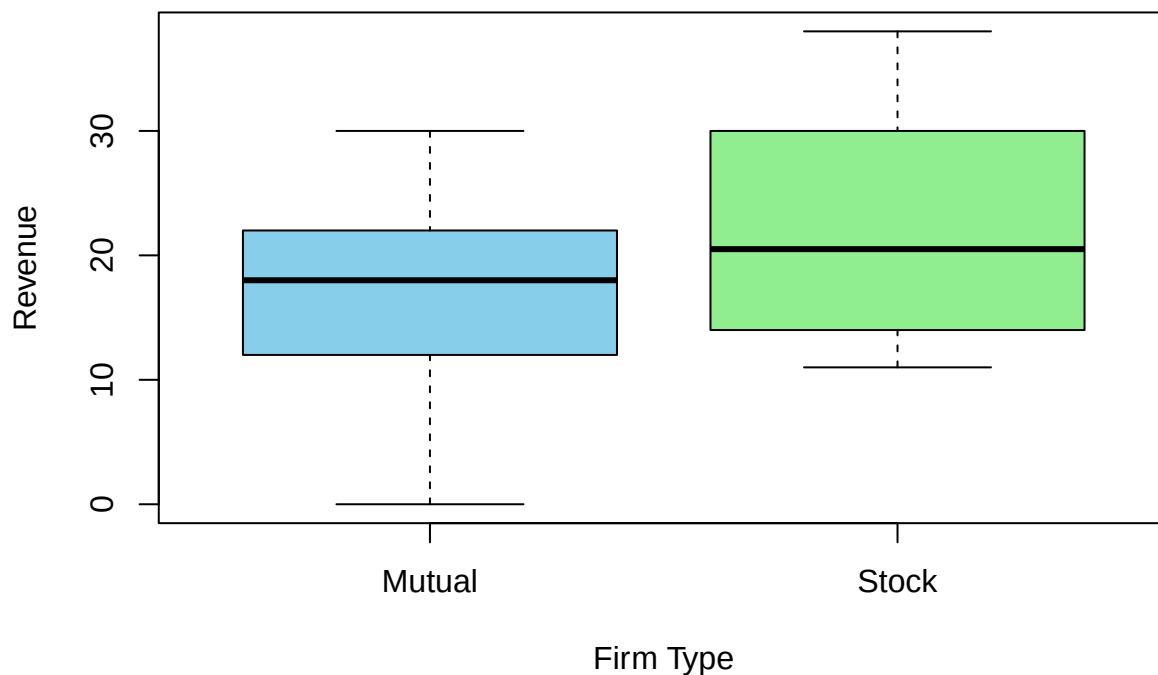
```
IID$X.Mutual=I(IID$Firm.Type=="Mutual")*1
IID$X.Mutual
```

```
## [1] 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0
```

```
f<-lm(Y~X1+X.Stock,data=IID)
summary(f)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X.Stock, data = IID)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.6915 -1.7036 -0.4385  1.9210  6.3406
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 33.874069   1.813858  18.675 9.15e-13 ***
## X1          -0.101742   0.008891 -11.443 2.07e-09 ***
## X.Stock       8.055469   1.459106   5.521 3.74e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.221 on 17 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8827
## F-statistic: 72.5 on 2 and 17 DF, p-value: 4.765e-09
# Create side-by-side boxplots
boxplot(IID$Y ~ IID$Firm.Type,
        col = c("skyblue", "lightgreen"),
        main = "Side-by-Side Boxplots",
        xlab = "Firm Type",
        ylab = "Revenue")
```

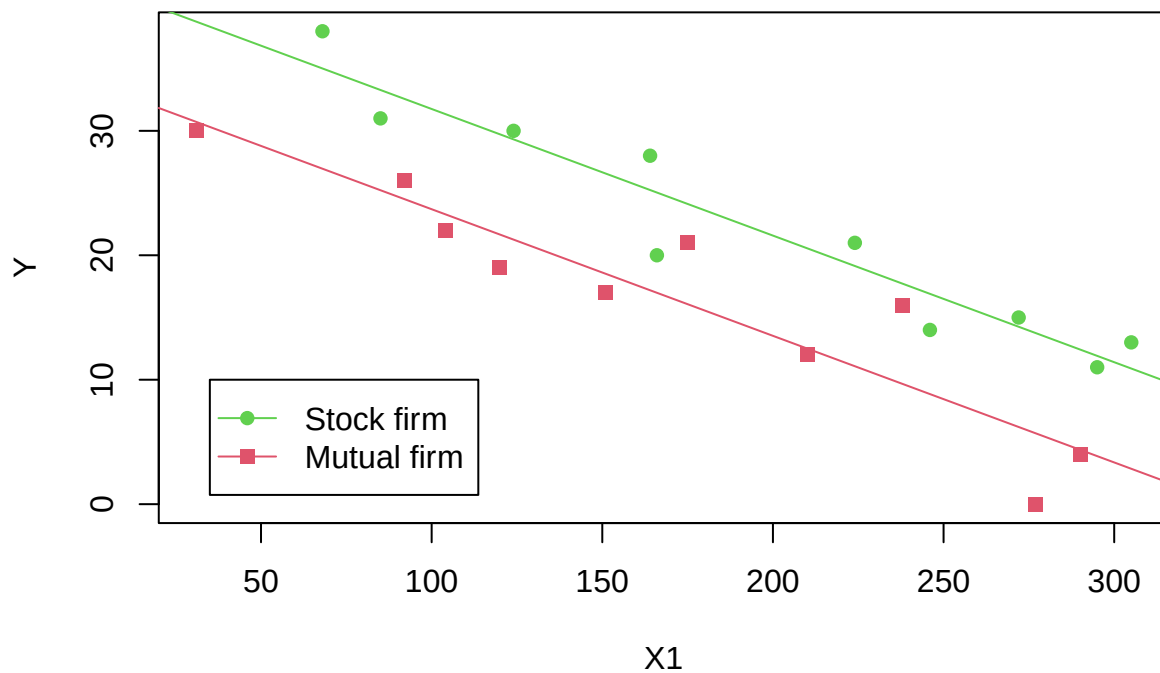
Side-by-Side Boxplots



Box plots overlap. Revenues for Mutual is ranged from 0 to 30. While, for stock, 10 to 40.

Regression Lines for Mutul and Stock Companies

```
Y=IID$Y
X1<-IID$X1
X2<-IID$X2
cX2<-as.factor(IID$X2)
fit<-lm(Y~X1 +cX2)
coef <- coefficients(fit)
plot( Y ~ X1, pch=(15+X2),col = as.character(2+X2))
abline(coef["(Intercept)"] + coef["cX21"], coef["X1"], col = 3)
abline(coef["(Intercept)"] , coef["X1"], col = 2)
legend(35,10,c("Stock firm","Mutual firm"),col=c(3,2), lwd=1,lty=c(1,1), pch=c(16,15))
```



the contrast matrix for categorical variable with two levels

```
contr.treatment(2)

##      2
## 1 0
## 2 1

IID$Firm.Type.f<-factor(IID$Firm.Type)
contrasts(IID$Firm.Type.f) = contr.treatment(2)
f<-lm(Y~X1+Firm.Type.f,data=IID)
summary(f)

##
## Call:
## lm(formula = Y ~ X1 + Firm.Type.f, data = IID)
##
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -5.6915 -1.7036 -0.4385  1.9210  6.3406
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  33.874069   1.813858  18.675 9.15e-13 ***
## X1          -0.101742   0.008891 -11.443 2.07e-09 ***
## Firm.Type.f2  8.055469   1.459106   5.521 3.74e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.221 on 17 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8827
## F-statistic: 72.5 on 2 and 17 DF,  p-value: 4.765e-09
```

Interaction Terms

```
#1st way interaction term
f<-lm(Y~X1+X2+X1*X2,data=IID)
summary(f)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X2 + X1 * X2, data = IID)
##
## Residuals:
##      Min      1Q  Median      3Q      Max
## -5.7144 -1.7064 -0.4557  1.9311  6.3259
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 33.8383695   2.4406498  13.864 2.47e-10 ***
## X1          -0.1015306   0.0130525  -7.779 7.97e-07 ***
## X2           8.1312501   3.6540517   2.225  0.0408 *
## X1:X2        -0.0004171   0.0183312  -0.023  0.9821
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.32 on 16 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8754
## F-statistic: 45.49 on 3 and 16 DF,  p-value: 4.675e-08
```

```
#2nd way interaction term
f<-lm(Y~X1+X2+X1:X2,data=IID)
summary(f)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X2 + X1:X2, data = IID)
##
## Residuals:
##      Min      1Q  Median      3Q      Max
## -5.7144 -1.7064 -0.4557  1.9311  6.3259
##
## Coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) 33.8383695  2.4406498  13.864 2.47e-10 ***
## X1          -0.1015306  0.0130525  -7.779 7.97e-07 ***
## X2           8.1312501  3.6540517   2.225  0.0408 *
## X1:X2        -0.0004171  0.0183312  -0.023  0.9821
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.32 on 16 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8754
## F-statistic: 45.49 on 3 and 16 DF,  p-value: 4.675e-08
```

#3rd way interaction term

```
IID$X3=IID$X1*IID$X2
f<-lm(Y~X1+X2+X3,data=IID)
summary(f)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X2 + X3, data = IID)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.7144 -1.7064 -0.4557  1.9311  6.3259
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) 33.8383695  2.4406498  13.864 2.47e-10 ***
## X1          -0.1015306  0.0130525  -7.779 7.97e-07 ***
## X2           8.1312501  3.6540517   2.225  0.0408 *
## X3          -0.0004171  0.0183312  -0.023  0.9821
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.32 on 16 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8754
## F-statistic: 45.49 on 3 and 16 DF,  p-value: 4.675e-08
```

#4th way

```
f1<-lm(Y~.^2,data=IID[,1:3])
summary(f1)
```

```
##
## Call:
## lm(formula = Y ~ .^2, data = IID[, 1:3])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.7144 -1.7064 -0.4557  1.9311  6.3259
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) 33.8383695  2.4406498  13.864 2.47e-10 ***
## X1          -0.1015306  0.0130525  -7.779 7.97e-07 ***
## X2           8.1312501  3.6540517   2.225  0.0408 *
## X1:X2        -0.0004171  0.0183312  -0.023  0.9821
```



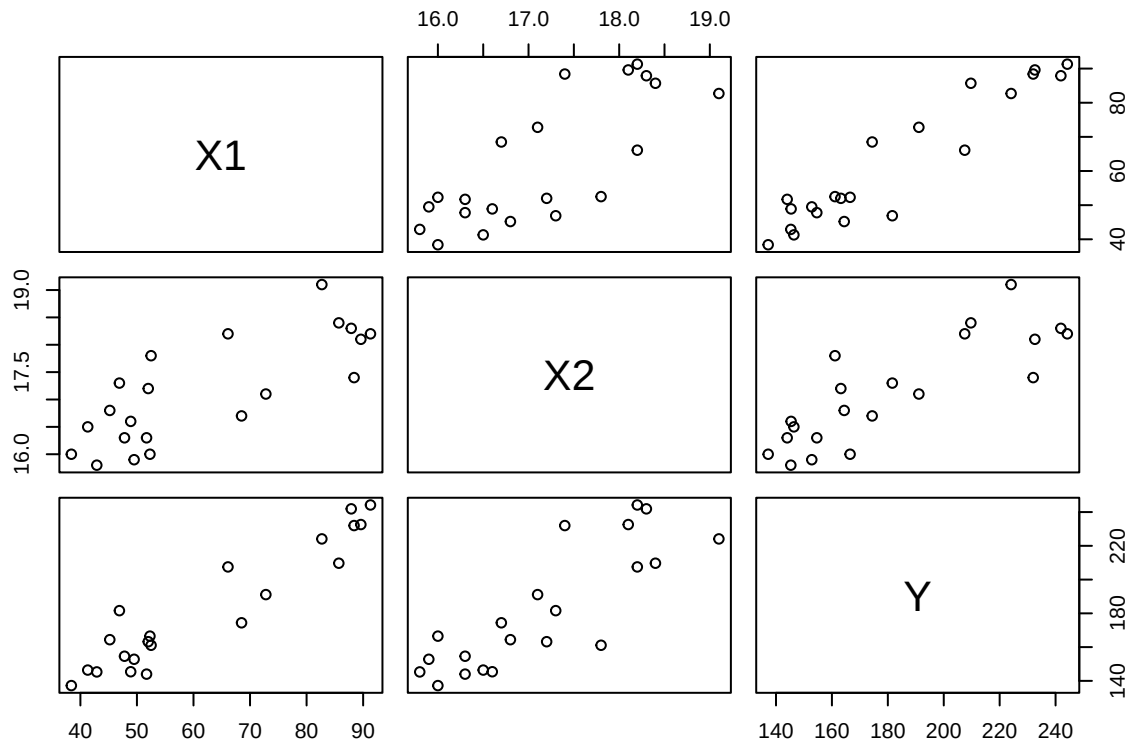
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.32 on 16 degrees of freedom
## Multiple R-squared:  0.8951, Adjusted R-squared:  0.8754
## F-statistic: 45.49 on 3 and 16 DF,  p-value: 4.675e-08
```

Example

```
DwaineStudio <- read.csv("/cloud/project/Data/DwaineStudio.csv")
round(cor(DwaineStudio),2)
```

```
##      X1  X2   Y
## X1 1.00 0.78 0.94
## X2 0.78 1.00 0.84
## Y   0.94 0.84 1.00
```

```
plot(DwaineStudio)
```



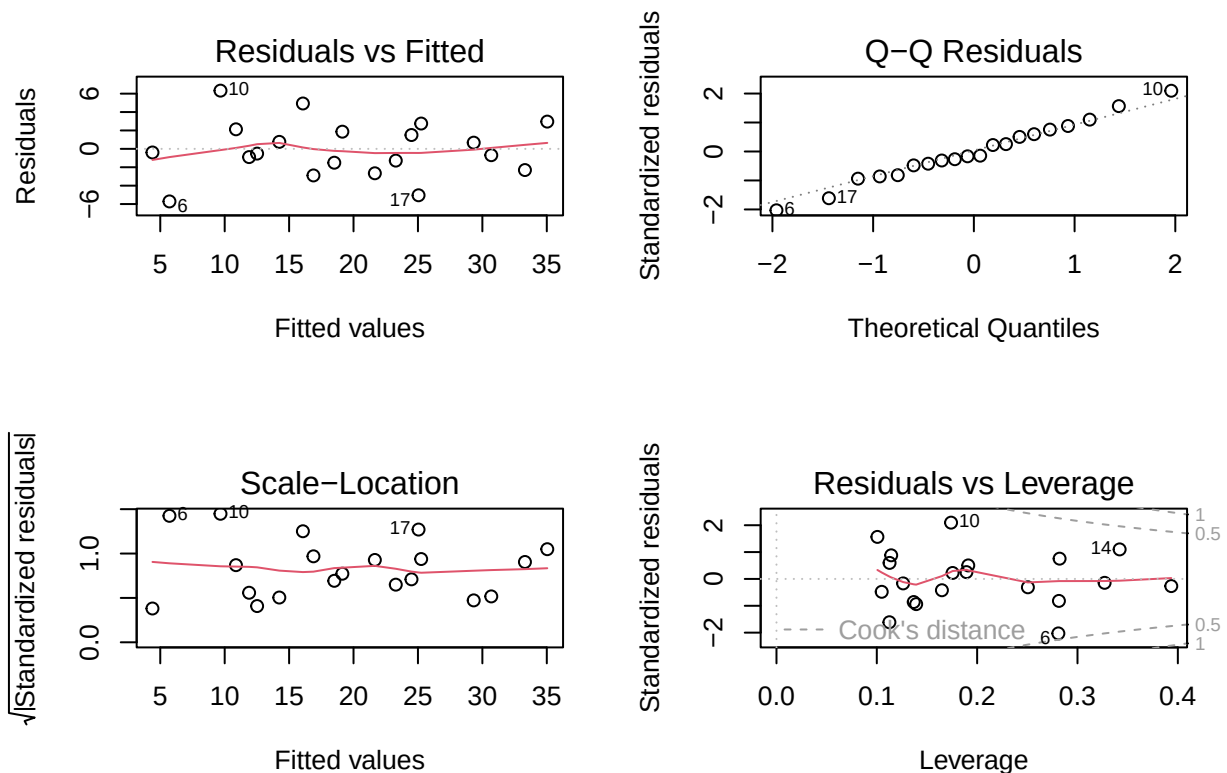
```
summary(DwaineStudio)
```

```
##      X1      X2      Y
## Min.   :38.40  Min.   :15.80  Min.   :137.2
## 1st Qu.:47.80  1st Qu.:16.30  1st Qu.:152.8
## Median :52.30  Median :17.10  Median :166.5
## Mean   :62.02  Mean   :17.14  Mean   :181.9
## 3rd Qu.:82.70  3rd Qu.:18.10  3rd Qu.:209.7
## Max.   :91.30  Max.   :19.10  Max.   :244.2
```

```
f.dw<-lm(Y~X1+X2,data=DwaineStudio )
summary(f.dw)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X2, data = DwaineStudio)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.4239  -6.2161   0.7449   9.4356  20.2151
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -68.8571    60.0170  -1.147  0.2663
## X1           1.4546     0.2118   6.868 2e-06 ***
## X2           9.3655     4.0640   2.305  0.0333 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.01 on 18 degrees of freedom
## Multiple R-squared:  0.9167, Adjusted R-squared:  0.9075
## F-statistic: 99.1 on 2 and 18 DF,  p-value: 1.921e-10
```

```
par(mfrow=c(2,2))
plot(f)
```



```
anova(f.dw)
```

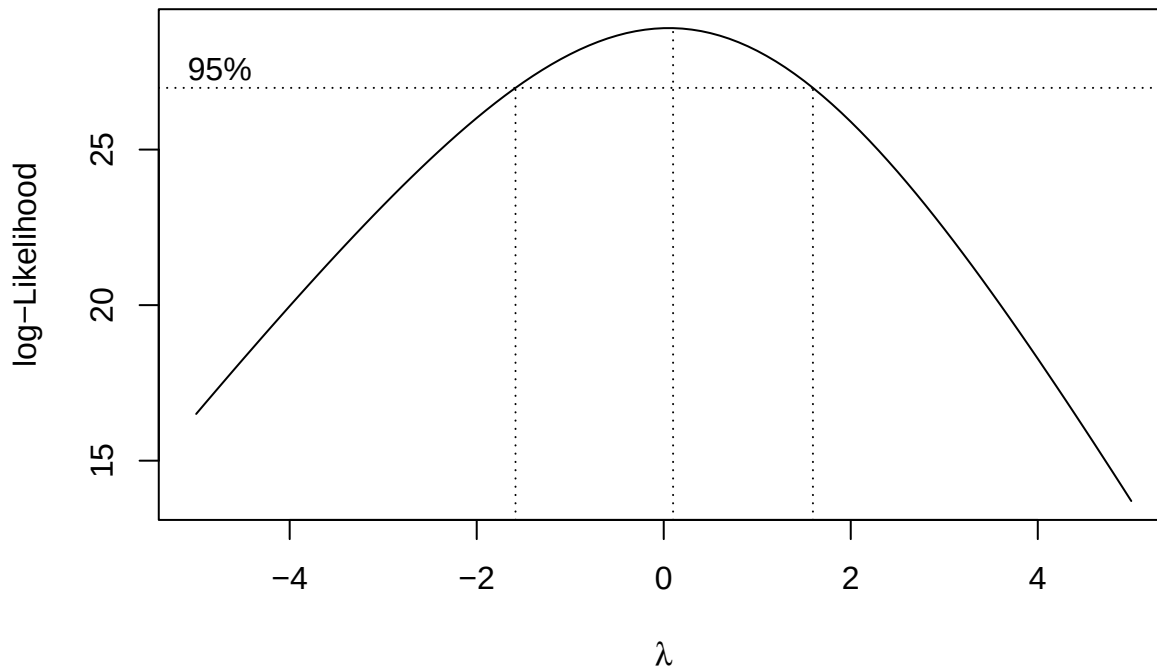
```
## Analysis of Variance Table
##
## Response: Y
##           Df Sum Sq Mean Sq F value Pr(>F)
## X1         1 23371.8 23371.8 192.8962 4.64e-11 ***
```

```
## X2          1    643.5    643.5    5.3108  0.03332 *
## Residuals 18  2180.9    121.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Box Cox Transformation #####
```

λ is very close to zero. To easy up the computation, we will use the log transformation.

```
library(MASS)
par(mfrow=c(1,1))
boxcox(f.dw,lambda=seq(-5,5,by=.1))
```



```
#getting the value of lambda
b=boxcox(f.dw,lambda=seq(-5,5,by=.1))
lambda<-b$x[which.max(b$y)]
lambda
```

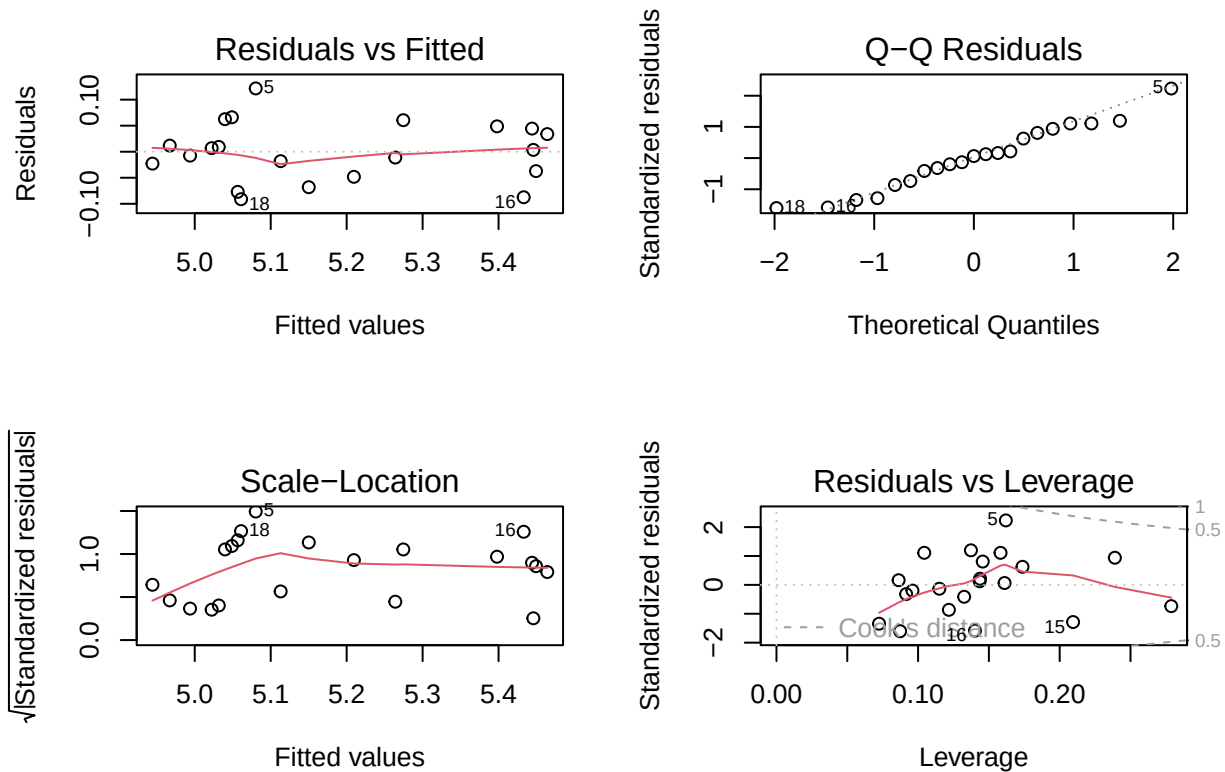
```
## [1] 0.1
```

```
#doing the log transformation
f1.dw<-lm(log(Y)~X1+X2,data=DwaineStudio )
summary(f1.dw)
```

```
##
## Call:
## lm(formula = log(Y) ~ X1 + X2, data = DwaineStudio)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.091173 -0.037237  0.003494  0.044405  0.121520
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.768914   0.324501  11.615 8.52e-10 ***
## X1           0.007523   0.001145   6.570 3.59e-06 ***
```

```
## X2          0.055408    0.021973    2.522    0.0213 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05951 on 18 degrees of freedom
## Multiple R-squared:  0.9149, Adjusted R-squared:  0.9055
## F-statistic: 96.78 on 2 and 18 DF,  p-value: 2.337e-10
```

```
#checking visually reg model assumptions
par(mfrow=c(2,2))
plot(f1.dw)
```



Correlation Test for Normality

H_o : Errors are normally distributed

H_a : Errors are Not normally distributed

```
shapiro.test(f1.dw$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  f1.dw$residuals
## W = 0.97305, p-value = 0.7992
```

Accept H_o , it is normally distributed

Brown-Forsythe Test

H_o : Error variances are constant

H_a : Error variances are not constant

We will use X1 to split the data in two parts

```
pop.dw<-data.frame(cbind(DwaineStudio$Y,DwaineStudio$X1, DwaineStudio$X2,e1=f1.dw$residuals))
library(car)
```

```
## Loading required package: carData
```

```
#Create the group variable based on medium X1 value (Low Xs =0 vs High Xs =1)
group=I(pop.dw[,2]< median(DwaineStudio$X1))*1
#using built in function by inputting errors ei=pop.dw[,4] and group variable as inputs
leveneTest(pop.dw[,4], group, center=median)
```

```
## Warning in leveneTest.default(pop.dw[, 4], group, center = median): group
## coerced to factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 1  0.0438 0.8364
##      19
```

P Value is greater than 0.05. Accept H_o , Error variances are constant.

Breusch-Pagan Test

H_o : Error variances are constant

H_a : Error variances are not constant

```
library(lmtest)
```

```
## Loading required package: zoo
```

```
##
```

```
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      as.Date, as.Date.numeric
```

```
bptest(log(Y)~X1+X2,data=DwaineStudio)
```

```
##
```

```
## studentized Breusch-Pagan test
```

```
##
```

```
## data: log(Y) ~ X1 + X2
```

```
## BP = 2.4088, df = 2, p-value = 0.2999
```

P Value is greater than 0.05. Accept H_o , Error variances are constant.

can we drop X2?

H_o : $\beta_2 = 0$

H_a : $\beta_2 \neq 0$

```
#model without X2
```

```
f2.dw<-lm(log(Y)~X1,data=DwaineStudio)
```

```
summary(f2.dw)
```

```
##
```

```
## Call:
```

```
## lm(formula = log(Y) ~ X1, data = DwaineStudio)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.114595 -0.033766 -0.005431  0.024720  0.164335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4.5788591  0.0522979   87.55 < 2e-16 ***
## X1          0.0097785  0.0008092   12.08 2.3e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06739 on 19 degrees of freedom
## Multiple R-squared:  0.8849, Adjusted R-squared:  0.8788
## F-statistic: 146 on 1 and 19 DF,  p-value: 2.304e-10
anova(f2.dw,f1.dw)

## Analysis of Variance Table
##
## Model 1: log(Y) ~ X1
## Model 2: log(Y) ~ X1 + X2
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      19 0.086279
## 2      18 0.063757  1  0.022523 6.3587 0.02133 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
p value is 0.0213 < 0.05, Reject  $H_o$ . We CAN NOT drop  $X_2$ 
```